

```
In [2]: l=[]
```

```
In [3]: l1=[10,45,6.5,'pragati',3,True,(2+3j)]  
l1
```

```
Out[3]: [10, 45, 6.5, 'pragati', 3, True, (2+3j)]
```

```
In [6]: l2=[10,20,30,40,50,10,30]
```

```
In [7]: print(l)  
print(l1)  
print(l2)
```

```
[]  
[10, 45, 6.5, 'pragati', 3, True, (2+3j)]  
[10, 20, 30, 40, 50, 10, 30]
```

## append()

```
l1.append(40) l1
```

## extend()

```
In [12]: l.extend(l2)  
l
```

```
Out[12]: [10, 20, 30, 40, 50, 10, 30]
```

```
In [13]: print(l)  
print(l2)
```

```
[10, 20, 30, 40, 50, 10, 30]  
[10, 20, 30, 40, 50, 10, 30]
```

```
In [15]: l1=[10,45,6.5,'pragati',3,True,(2+3j)]  
l1
```

```
Out[15]: [10, 45, 6.5, 'pragati', 3, True, (2+3j)]
```

```
In [16]: l4=[2,3,4]  
l4
```

```
Out[16]: [2, 3, 4]
```

## extend()

```
In [18]: l4.extend(l1)  
l4
```

```
Out[18]: [2,
          3,
          4,
          10,
          45,
          6.5,
          'pragati',
          3,
          True,
          (2+3j),
          10,
          45,
          6.5,
          'pragati',
          3,
          True,
          (2+3j)]
```

## clear( list no.)

```
In [21]: clear(l3)           # total clear l3
```

```
In [22]: l2
```

```
Out[22]: [10, 20, 30, 40, 50, 10, 30]
```

```
In [27]: l5=[3,4,5,6]
          l5
```

```
Out[27]: [3, 4, 5, 6]
```

## copy()

```
In [32]: l6=l5.copy()
          l6
```

```
Out[32]: [3, 4, 5, 6]
```

```
In [33]: l1
```

```
Out[33]: [10, 20, 30, 40, 50, 10, 30]
```

## index(value)

```
In [34]: l.index(30)         # 1st occurence
```

```
Out[34]: 2
```

```
In [35]: l1
```

```
Out[35]: [10, 20, 30, 40, 50, 10, 30]
```

```
In [40]: l1=[10, 20, 30, 40, 50, 10, 30]  
l1
```

```
Out[40]: [10, 20, 30, 40, 50, 10, 30]
```

## insert(index,value)

```
In [41]: l1.insert(3,37)      # insert(index,value)  
l1
```

```
Out[41]: [10, 20, 30, 37, 40, 50, 10, 30]
```

## pop()

pop() --> remove last value pop(index) --> remove that specific index value

```
In [44]: l1
```

```
Out[44]: [10, 20, 30, 37, 40, 50, 10, 30]
```

```
In [45]: l1.pop()           # Last value remove  
l1
```

```
Out[45]: [10, 20, 30, 37, 40, 50, 10]
```

```
In [46]: l1
```

```
Out[46]: [10, 20, 30, 37, 40, 50, 10]
```

```
In [47]: l1.pop(3)          # remove 3rd index value  
l1
```

```
Out[47]: [10, 20, 30, 40, 50, 10]
```

## remove(value)

```
In [48]: l1
```

```
Out[48]: [10, 20, 30, 40, 50, 10]
```

```
In [49]: l1.remove(40)  
l1
```

```
Out[49]: [10, 20, 30, 50, 10]
```

## reverse()

```
In [50]: l2=[2,4,5,7,5]
12
```

```
Out[50]: [2, 4, 5, 7, 5]
```

```
In [51]: l2.reverse()
12
```

```
Out[51]: [5, 7, 5, 4, 2]
```

## sort

sort() --> ascending sort(reverse=True) --> descending

```
In [54]: l4=[2,5,3,6,84,26]
14
```

```
Out[54]: [2, 5, 3, 6, 84, 26]
```

```
In [55]: l4.sort()
14
```

```
Out[55]: [2, 3, 5, 6, 26, 84]
```

```
In [56]: l4=[3,5,13,21,643,87]
14
```

```
Out[56]: [3, 5, 13, 21, 643, 87]
```

```
In [57]: l4.sort(reverse=True)
14
```

```
Out[57]: [643, 87, 21, 13, 5, 3]
```

## tunning

- parameter tuning --> inbuild sysyem parameter - hyperparameter tuning --> user change the system parameter

```
In [58]: l=[3,4,2,7,4,84,3]
1
```

```
Out[58]: [3, 4, 2, 7, 4, 84, 3]
```

```
In [60]: l.sort()           # parameter tunning bcoz sort is inbuild fun
                        # in original (reverse=False)
1
```

```
Out[60]: [2, 3, 3, 4, 4, 7, 84]
```

```
In [62]: l1=[2,4,112,6,4,20,90]
11
```

```
Out[62]: [2, 4, 112, 6, 4, 20, 90]
```

```
In [63]: l1.sort(reverse=True)      # hyperparameter tunning
        l1                        # original (reverse=False)
```

```
Out[63]: [112, 90, 20, 6, 4, 4, 2]
```

## double indexing

```
In [65]: l3=[2,5,5.7,10,'pragati',(3+5j),True]
        l3
```

```
Out[65]: [2, 5, 5.7, 10, 'pragati', (3+5j), True]
```

```
In [66]: l3[2]
```

```
Out[66]: 5.7
```

```
In [67]: l3[4]
```

```
Out[67]: 'pragati'
```

```
In [68]: l3[4][3]      # p r a g a t i
```

```
Out[68]: 'g'
```

```
In [ ]: # how to print a name in vertical
```

```
In [70]: print(l3[4][0])
        print(l3[4][1])
        print(l3[4][2])
        print(l3[4][3])
        print(l3[4][4])
        print(l3[4][5])
        print(l3[4][6])
```

p  
r  
a  
g  
a  
t  
i

```
In [ ]:
```